



Computing Infrastructures
January 13, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

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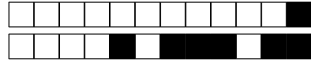
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.

**True false questions**

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 SANs are primarily used for block-level access to data, while NAS devices provide file-level access.

☐ A False

☐ B True

Question 2 RAID can be used with both hardware and software implementations.

☐ A False

☐ B True

Question 3 GPUs (Graphics Processing Units) are primarily used in datacenters for accelerating graphical applications and gaming.

☐ A False

☐ B True

Question 4 Cloud computing applications do not require any server to run.

☐ A True

☐ B False

Question 5 Availability zones within a Compute Region can have a roundtrip time larger than few milli seconds.

☐ A False

☐ B True

Question 6 Edge computing is a distributed computing paradigm that brings computation and data storage closer to the location where it is needed.

☐ A False

☐ B True

Question 7 The power consumption of an average datacenter is in the order of a few GigaWatts.

☐ A False

☐ B True

Question 8 Leaf-spine topologies are more suitable for small-scale datacenter deployments with limited server connections.

☐ A False

☐ B True

Question 9 Warehouse-scale computers are typically composed of a large set of homogeneous servers.

☐ A False

☐ B True

Question 10 Virtualization can help simplify IT management by reducing the number of physical machines that need to be managed.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

A scientific computation that needs to be carried out within the PoliMi data center uses a server composed of 2 CPUs and 4 GPUs. Knowing that:

- The computation takes 8 days to complete,
- The computation requires at least one CPU and all GPUs within the server to be operational to complete successfully;
- $MTTF_{CPU} = 180$ days and $MTTF_{GPU} = 120$ days.

How many parallel instances of the computation must be launched to ensure a probability higher than 98% that at least one computation produces results successfully? Notes: (i) Use at least 4 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.

Question 12

If the time required to recover and replace a component of the server described in the previous exercise (whether it is a CPU or a GPU) is equal to 24 days, what is the total availability of the server? Notes: (i) Use at least 5 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.



+1/4/57+

Question 13

A company is planning to use a RAID 5 array composed of 8 disks for critical data storage. The desired Mean Time to Failure for the entire RAID system ($MTTF_{RAID5}$) is 16 years. Each disk in the array has a $MTTF_{disk}$ equal to 500 days. What should be the MTTR required *in hours* to meet the target RAID $MTTF_{RAID5}$.

Question 14

Consider a system composed of three stations: the CPU that is characterized by $V_{CPU} = 100$ visits and an average service time of $S_{CPU} = 10\text{ms}$; the disk, characterized by a throughput of 12 IOPS, and a demand of $D_{DISK} = 150\text{ms}$; and the GPU whose demand is $D_{GPU} = 40\text{s}$ and the number of visits $V_{GPU} = 10$. Finally, the system throughput is $X = 20$ jobs/min while the response time when there are $N = 20$ end-users in the system is $R = 55\text{s}$.

Compute the CPU demand and the GPU throughput.

Write in the answer sheet: $D_{CPU} = \dots$; $X_{GPU} = \dots$



+1/5/56+

Question 15

Considering the system described in Question 14, what is the users' think time Z ?

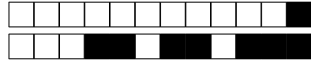
Question 16

Based on the system in Question 14, the number of end users is predicted to reach 40 in one month. Considering the response time lower bound, which option is better?

- Upgrade the system by adding one more GPU (you can assume the new GPU is equal to the one initially available and to balance evenly the GPU processing).
- Replace the GPU with one 2.5 times faster than the original one.

Provide the estimated bounds in the two scenarios to motivate your answer.

Write in the answer sheet: A or B ; $R_{LOW}^A = \dots$; $R_{LOW}^B = \dots$



+1/6/55+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

⇒ What are the advantages and disadvantages of using Type 1 hypervisors versus Type 2 hypervisors in a virtualized environment?

Question 18

⇒ What are the implications of hardware heterogeneity on software stack development in datacenters?

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Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

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Question 10 : ☐A ☐B

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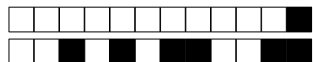
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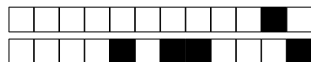
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True false questions

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Question 1 Warehouse-scale computers are typically composed of a large set of homogeneous servers.

☐ A True

☐ B False

Question 2 Edge computing is a distributed computing paradigm that brings computation and data storage closer to the location where it is needed.

☐ A True

☐ B False

Question 3 Availability zones within a Compute Region can have a roundtrip time larger than few milli seconds.

☐ A False

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Question 4 SANs are primarily used for block-level access to data, while NAS devices provide file-level access.

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Question 5 Leaf-spine topologies are more suitable for small-scale datacenter deployments with limited server connections.

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Question 6 GPUs (Graphics Processing Units) are primarily used in datacenters for accelerating graphical applications and gaming.

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Question 9 Virtualization can help simplify IT management by reducing the number of physical machines that need to be managed.

☐ A False

☐ B True

Question 10 Cloud computing applications do not require any server to run.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

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A scientific computation that needs to be carried out within the PoliMi data center uses a server composed of 2 CPUs and 2 GPUs. Knowing that:

- The computation takes 8 days to complete,
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Question 13

A company is planning to use a RAID 5 array composed of 8 disks for critical data storage. The desired Mean Time to Failure for the entire RAID system ($MTTF_{RAID5}$) is 12 years. Each disk in the array has a $MTTF_{disk}$ equal to 500 days. What should be the MTTR required *in hours* to meet the target RAID $MTTF_{RAID5}$.

Question 14

Consider a system composed of three stations: the CPU that is characterized by $V_{CPU} = 50$ visits and an average service time of $S_{CPU} = 10\text{ms}$; the disk, characterized by a throughput of 10 IOPS, and a demand of $D_{DISK} = 300\text{ms}$; and the GPU whose demand is $D_{GPU} = 30\text{s}$ and the number of visits $V_{GPU} = 15$. Finally, the system throughput is $X = 25$ jobs/min while the response time when there are $N = 20$ end-users in the system is $R = 38\text{s}$.

Compute the CPU demand and the GPU throughput.

Write in the answer sheet: $D_{CPU} = \dots; X_{GPU} = \dots$



+2/5/46+

Question 15

Considering the system described in Question 14, what is the users' think time Z ?

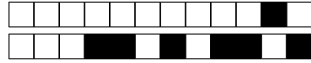
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+2/6/45+

Open Questions

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Question 17

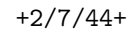
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Exercises

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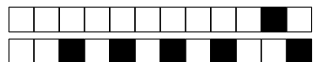
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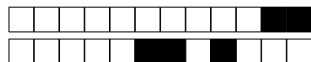
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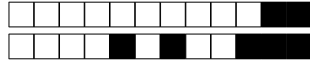
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☐ A False

☐ B True

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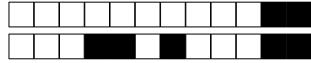
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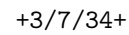
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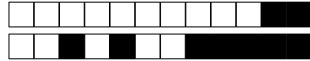
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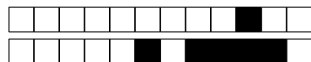
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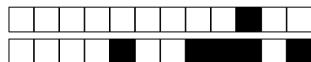
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True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Cloud computing applications do not require any server to run.

☐ A True

☐ B False

Question 2 Warehouse-scale computers are typically composed of a large set of homogeneous servers.

☐ A False

☐ B True

Question 3 Availability zones within a Compute Region can have a roundtrip time larger than few milli seconds.

☐ A True

☐ B False

Question 4 Edge computing is a distributed computing paradigm that brings computation and data storage closer to the location where it is needed.

☐ A False

☐ B True

Question 5 SANs are primarily used for block-level access to data, while NAS devices provide file-level access.

☐ A True

☐ B False

Question 6 The power consumption of an average datacenter is in the order of a few GigaWatts.

☐ A True

☐ B False

Question 7 Leaf-spine topologies are more suitable for small-scale datacenter deployments with limited server connections.

☐ A False

☐ B True

Question 8 RAID can be used with both hardware and software implementations.

☐ A False

☐ B True

Question 9 Virtualization can help simplify IT management by reducing the number of physical machines that need to be managed.

☐ A False

☐ B True

Question 10 GPUs (Graphics Processing Units) are primarily used in datacenters for accelerating graphical applications and gaming.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

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How many parallel instances of the computation must be launched to ensure a probability higher than 98% that at least one computation produces results successfully? Notes: (i) Use at least 4 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.

Question 12

If the time required to recover and replace a component of the server described in the previous exercise (whether it is a CPU or a GPU) is equal to 24 days, what is the total availability of the server? Notes: (i) Use at least 5 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.



Question 13

A company is planning to use a RAID 5 array composed of 8 disks for critical data storage. The desired Mean Time to Failure for the entire RAID system ($MTTF_{RAID5}$) is 12 years. Each disk in the array has a $MTTF_{disk}$ equal to 500 days. What should be the MTTR required *in hours* to meet the target RAID $MTTF_{RAID5}$.

Question 14

Consider a system composed of three stations: the CPU that is characterized by $V_{CPU} = 100$ visits and an average service time of $S_{CPU} = 10\text{ms}$; the disk, characterized by a throughput of 12 IOPS, and a demand of $D_{DISK} = 150\text{ms}$; and the GPU whose demand is $D_{GPU} = 40\text{s}$ and the number of visits $V_{GPU} = 10$. Finally, the system throughput is $X = 20$ jobs/min while the response time when there are $N = 20$ end-users in the system is $R = 55\text{s}$.

Compute the CPU demand and the GPU throughput.

Write in the answer sheet: $D_{CPU} = \dots; X_{GPU} = \dots$



+4/5/26+

Question 15

Considering the system described in Question 14, what is the users' think time Z ?

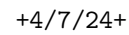
Question 16

Based on the system in Question 14, the number of end users is predicted to reach 40 in one month. Considering the response time lower bound, which option is better?

- Upgrade the system by adding one more GPU (you can assume the new GPU is equal to the one initially available and to balance evenly the GPU processing).
- Replace the GPU with one 2.5 times faster than the original one.

Provide the estimated bounds in the two scenarios to motivate your answer.

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Answer Sheets (Page 1)

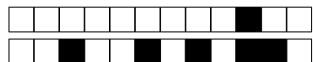
Student ID (Codice Persona):

⇒ What are the advantages and disadvantages of using Type 1 hypervisors versus Type 2 hypervisors in a virtualized environment?

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Computing Infrastructures - January 13, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

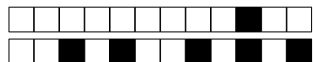
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



+4/10/21+

Computing Infrastructures
January 13, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

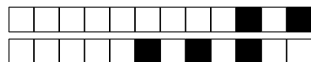
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

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If needed, you can use this page for notes. Any answer written here will be ignored.



+5/1/20+

Computing Infrastructures

January 13, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
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Exam Duration: 1hour and 30min

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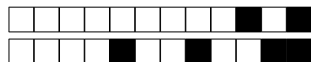
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True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

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Question 1 Edge computing is a distributed computing paradigm that brings computation and data storage closer to the location where it is needed.

☐ A True

☐ B False

Question 2 Cloud computing applications do not require any server to run.

☐ A True

☐ B False

Question 3 GPUs (Graphics Processing Units) are primarily used in datacenters for accelerating graphical applications and gaming.

☐ A True

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☐ A True

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☐ B False

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☐ A False

☐ B True

Question 8 Virtualization can help simplify IT management by reducing the number of physical machines that need to be managed.

☐ A True

☐ B False

Question 9 Warehouse-scale computers are typically composed of a large set of homogeneous servers.

☐ A False

☐ B True

Question 10 The power consumption of an average datacenter is in the order of a few Gi-gaWatts.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

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How many parallel instances of the computation must be launched to ensure a probability higher than 98% that at least one computation produces results successfully? Notes: (i) Use at least 4 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.

Question 12

If the time required to recover and replace a component of the server described in the previous exercise (whether it is a CPU or a GPU) is equal to 24 days, what is the total availability of the server? Notes: (i) Use at least 5 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.



+5/4/17+

Question 13

A company is planning to use a RAID 5 array composed of 8 disks for critical data storage. The desired Mean Time to Failure for the entire RAID system ($MTTF_{RAID5}$) is 16 years. Each disk in the array has a $MTTF_{disk}$ equal to 500 days. What should be the MTTR required *in hours* to meet the target RAID $MTTF_{RAID5}$.

Question 14

Consider a system composed of three stations: the CPU that is characterized by $V_{CPU} = 100$ visits and an average service time of $S_{CPU} = 10\text{ms}$; the disk, characterized by a throughput of 12 IOPS, and a demand of $D_{DISK} = 150\text{ms}$; and the GPU whose demand is $D_{GPU} = 40\text{s}$ and the number of visits $V_{GPU} = 10$. Finally, the system throughput is $X = 20$ jobs/min while the response time when there are $N = 20$ end-users in the system is $R = 55\text{s}$.

Compute the CPU demand and the GPU throughput.

Write in the answer sheet: $D_{CPU} = \dots; X_{GPU} = \dots$



+5/5/16+

Question 15

Considering the system described in Question 14, what is the users' think time Z ?

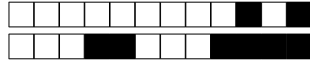
Question 16

Based on the system in Question 14, the number of end users is predicted to reach 40 in one month. Considering the response time lower bound, which option is better?

- Upgrade the system by adding one more GPU (you can assume the new GPU is equal to the one initially available and to balance evenly the GPU processing).
- Replace the GPU with one 2.5 times faster than the original one.

Provide the estimated bounds in the two scenarios to motivate your answer.

Write in the answer sheet: A or B ; $R_{LOW}^A = \dots$; $R_{LOW}^B = \dots$



+5/6/15+

Open Questions

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Question 17

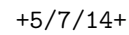
⇒ What are the advantages and disadvantages of using Type 1 hypervisors versus Type 2 hypervisors in a virtualized environment?

Question 18

⇒ What are the implications of hardware heterogeneity on software stack development in datacenters?

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Answer Sheets (Page 1)

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Computing Infrastructures - January 13, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

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Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

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Question 10 : ☐A ☐B

Exercises

Question 11 :

Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



Computing Infrastructures
January 13, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

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First Name:
(FIRST NAME IN CAPITAL LETTERS)

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+6/1/10+

Computing Infrastructures

January 13, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
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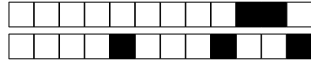
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☐ A True

☐ B False

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☐ A True

☐ B False

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☐ A True

☐ B False

Question 4 Cloud computing applications do not require any server to run.

☐ A True

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Question 6 The power consumption of an average datacenter is in the order of a few GigaWatts.

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+6/4/7+

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Question 14

Consider a system composed of three stations: the CPU that is characterized by $V_{CPU} = 50$ visits and an average service time of $S_{CPU} = 10\text{ms}$; the disk, characterized by a throughput of 10 IOPS, and a demand of $D_{DISK} = 300\text{ms}$; and the GPU whose demand is $D_{GPU} = 30\text{s}$ and the number of visits $V_{GPU} = 15$. Finally, the system throughput is $X = 25$ jobs/min while the response time when there are $N = 20$ end-users in the system is $R = 38\text{s}$.

Compute the CPU demand and the GPU throughput.

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+6/5/6+

Question 15

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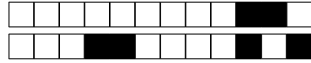
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+6/6/5+

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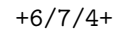
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Answer Sheets (Page 1)

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[illegible]



●



Computing Infrastructures - January 13, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



+6/10/1+

Computing Infrastructures
January 13, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

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